Optimal Single-Track Gray Codes from Gray-Ordered Necklaces

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Abstract

A single-track Gray code of length n is a cyclic Gray code of n-bit words in which every column is a cyclic shift of the first. For prime n the longest possible period is $2^n - 2$. Whether it is attained for every prime has been open since Hiltgen, Paterson and Brandestini introduced these codes in 1996, and optimal codes were known for primes $n \leq 19$. We show that a conjecture of Degni and Drisko (2007) on the reflected Gray code order of necklaces implies optimal single-track Gray codes for every prime. Their list of necklaces, conjecturally a Gray path for prime n, closes into a Hamilton cycle of the necklace graph after $(n-3)/2$ explicit Pósa rotations, and such a cycle lifts to an optimal code. This implication is formalized in Lean 4. Together with computation it gives optimal single-track Gray codes for every odd prime $n \leq 47$: for $n \leq 37$ explicit codes checked word by word by a program independent of the generator, and for $n = 41$, 43 and 47 codes obtained from the implication after the Degni–Drisko property was verified at these lengths by two independently written programs.

Index Terms

Single-track Gray codes, necklaces, reflected Gray code, Hamilton cycles, absolute rotary encoders, formal verification, Lean.

I. Introduction

A *single-track Gray code* of length n and period P is a cyclic sequence $w_0, \ldots, w_{P-1}$ of distinct n-bit words in which consecutive words (cyclically) differ in exactly one bit, and in which every column is a cyclic shift of column 0: there are *head positions* $h_0 = 0, h_1, \ldots, h_{n-1}$ with $(w_t)_j = (w_{t+h_j})_0$ for all t and j (indices mod P). Hiltgen, Paterson and Brandestini [6] introduced these codes for absolute rotary encoders. They showed that for $n \geq 3$ such a code contains neither 0^n nor 1^n and that $2n$ divides P. For prime n the period is therefore at most $2^n - 2$, and a code of period $2^n - 2$ is called *optimal*. For n a power of 2, Etzion and Paterson [2] constructed codes of period $2^n - 2n$, and Schwartz and Etzion [12] showed that this is best possible.

For prime n, optimal codes were known for $n \leq 19$. Mütze's survey [9, Section 3.12] records that existence "has been tested and confirmed for all prime $n \leq 19$" and lists the general question as open; Etzion [3] also lists the primes up to 19 as the known cases. Zhang and Zhu [13] list $2^n - 2$ as the optimal period with evenly spread heads for all primes $n \leq 29$, derived from counting arguments (the number of aperiodic necklaces and a necessary condition on the sequence of weights); they report codes "with length up to 17" and print none. We know of no published optimal single-track Gray code for a prime $n \geq 23$.

Mütze [9, Section 3.12] describes the standard reduction. For prime n every nonconstant necklace has n distinct rotations, and an optimal code arises from a cyclic Gray code through the $(2^n - 2)/n$ nonconstant necklaces, one representative each, in which the last representative is one flip away from a *nontrivial* rotation of the first. Lemma 1 shows that the nontriviality condition can always be met. The problem is therefore exactly the Hamiltonicity of the *necklace graph* Q_n, whose vertices are the nonconstant necklaces of length n and whose edges join necklaces having representatives at Hamming distance one.

Degni and Drisko [1] order necklaces by the first occurrence of a representative in the binary reflected Gray code (BRGC). They conjectured, and verified for $n \leq 37$, that consecutive necklaces in this order differ in a single bit except immediately after powers of odd primes (in their sense). For prime n this says that their list is a Gray path through all necklaces, from 0^n to 1^n; this is the prime case of Savage's question whether necklaces of odd length admit a Gray code [11], [9, Problem P17]. Without its constant ends the path runs from $\langle 0^{n-1}1 \rangle$ to $\langle 01^{n-1} \rangle$, which are not adjacent for $n \geq 5$. We close it explicitly.

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Theorem A: Let $n \geq 7$ be prime. If the Degni–Drisko list of necklaces of length n is a Gray path, then Q_n has a Hamilton cycle, obtained from the list by $(n-3)/2$ Pósa rotations at explicit pivots, and an optimal single-track Gray code of length n and period $2^n - 2$ exists.

Theorem B: Optimal single-track Gray codes exist for every odd prime $n \leq 47$. For $n \leq 37$ the codes are given explicitly and each has been checked word by word by a verifier independent of the generator (for $n = 3$, Lean's kernel); for $n = 41$, 43 and 47 the codes are the ones produced by Theorem A, and the hypothesis of Theorem A was verified at these lengths by two independently written programs.

Theorem A is formalized in Lean 4 with Mathlib (supplementary material). The Lean theorem `stgc_of_DD` states that for every prime $n \neq 2$ the Degni–Drisko property at n, an explicit hypothesis, implies the existence of a single-track Gray code of length n and period $2^n - 2$; its proof uses only Lean's standard axioms. The hypothesis there, adjacency in Q_n of consecutive necklaces of D_n, is Conjecture 1 below. It is implied by the conjecture as stated in [1], which asks that consecutive first-occurrence words, constants included, differ in one bit. Lean proves the hypothesis itself only for $n = 5$, by computation. For $7 \leq n \leq 37$ it was verified by Degni and Drisko [1], and for $n = 37$, 41, 43 and 47 by the computations of Section IV.

II. From Hamilton cycles in the necklace graph to codes

Words are $x = x_1 x_2 \cdots x_n$, and $\rho(x) = x_2 \cdots x_n x_1$ is the left rotation; $\langle x \rangle$ denotes the necklace (rotation class) of x, and e_i the word with a single 1 in position i.

Lemma 1: Let n be prime. If Q_n has a Hamilton cycle, then there is an optimal single-track Gray code of length n, with equally spaced heads.

Proof: Let $N_0, \ldots, N_{m-1}$ be the cycle, $m = (2^n - 2)/n$. Choose $w_0 \in N_0$ and, inductively, bits b_i with $w_{i+1} = w_i \oplus e_{b_i} \in N_{i+1}$ for $i < m-1$; finally choose b_{m-1} with $w_{m-1} \oplus e_{b_{m-1}} \in N_0$, say $w_{m-1} \oplus e_{b_{m-1}} = \rho^s(w_0)$. If $s \not\equiv 0 \pmod n$, put $W_{qm+i} = \rho^{qs}(w_i)$ for $0 \leq q < n$, $0 \leq i < m$. Rotations are automorphisms of the cube, so consecutive words differ in one bit, also across the boundaries $W_{qm+m-1} \to W_{(q+1)m} = \rho^{(q+1)s}(w_0)$ and cyclically since $ns \equiv 0$. The word W_{qm+i} lies in N_i, and $q \mapsto \rho^{qs}(w_i)$ is injective because s is invertible mod n and N_i has n distinct rotations; so the $nm = 2^n - 2$ words are exactly the nonconstant words. Finally $W_{t+m} = \rho^s(W_t)$ for all t, so column $ks \bmod n$ is column 0 delayed by km, which is the single-track property with heads $h_{ks} = -km \bmod (2^n - 2)$.

It remains to make $s \neq 0$. Rotate the cycle so that $N_0 = \langle 0^{n-1}1 \rangle$ and take $w_0 = e_a$, a word with a single 1, in position a; then $w_1 = e_a \oplus e_b$ for some $b \neq a$. Identify a word with the set of positions of its ones, so that ρ shifts every position down by one (mod n), and put $i \equiv b - a \not\equiv 0 \pmod n$. Replace the lift by $w'_0 = w_0$ and $w'_j = \rho^i(w_j)$ for $1 \leq j < m$. Since $\rho^i(w_1) = \{a - i, b - i\} = \{2a - b, a\}$ contains a, the first step $w'_0 \to w'_1$ is still a single flip; the later steps are single flips because ρ^i is a cube automorphism; $w'_j \in N_j$ for all j; and the closing step now ends at $\rho^i(\rho^s(w_0)) = \rho^{s+i}(w_0)$. So the new shift is $s + i$, and since $i \not\equiv 0$, one of s and $s + i$ is nonzero. ∎

III. The Degni–Drisko list and its closing

The BRGC lists $g(t) = t \oplus \lfloor t/2 \rfloor$, $t = 0, \ldots, 2^n - 1$, as n-bit words with the most significant bit first. Its inverse is the prefix-parity map: $\mathrm{rk}(x) = g^{-1}(x)$ has binary digits $\pi(x)_i = x_1 \oplus \cdots \oplus x_i$. Hence for words of equal length, $\mathrm{rk}(x) < \mathrm{rk}(y)$ if and only if $\pi(x)$ is lexicographically smaller than $\pi(y)$. The *first occurrence* of a necklace is its rotation of smallest rank. Let D_n be the list of the nonconstant necklaces of length n ordered by the ranks of their first occurrences. (Degni and Drisko list all necklaces; 0^n has rank 0 and 1^n is the only necklace whose first occurrence begins with 1, so D_n is a contiguous part of their list.)

Conjecture 1 (Degni–Drisko [1], prime case): For prime n, consecutive necklaces of D_n are adjacent in Q_n.

Degni and Drisko verified their conjecture for $n \leq 37$ [1, Section 3]. For composite odd n the adjacency fails, for example at $n = 9$, 15 and 21, as the exceptions in their conjecture allow.

Let $n \geq 7$ be odd and $K = (n-3)/2$. Define the words

$$E_0 = 0\,1^{n-1}, \qquad A_0 = 0\,1^{n-5}\,0111, \qquad E_1 = 0\,1^{n-5}\,0101,$$

and for $1 \leq k \leq K - 1$

$$A_k = 0^{2k+1}\,1\,0\,1^{n-2k-3}, \qquad E_{k+1} = 0^{2k+1}\,1\,0\,1^{n-2k-5}\,0\,1.$$

Thus $E_K = 0^{n-4}1001$. Write $c_k = n-2k-3$, which is even, with $c_k \geq 2$ for $k \leq K-1$ and $c_{K-1} = 2$. A *Pósa rotation* of a path $P = (P_0, \ldots, P_L)$ at a pivot P_j adjacent to the end P_L replaces P by $P_0 \cdots P_j\, P_L P_{L-1} \cdots P_{j+1}$, whose new end is P_{j+1}.

Lemma 2: For odd n the first necklace of D_n is $\langle 0^{n-1}1\rangle$ and the last is $\langle E_0\rangle$.

Proof: The first claim holds because $g(1) = 0^{n-1}1$. The first occurrence of $\langle E_0\rangle$ is E_0 itself (rotations beginning with 1 have larger rank), with $\pi(E_0) = 0(10)^{(n-1)/2}$. Let $X \neq \langle E_0\rangle$ be a nonconstant necklace; it has at least two zeros. Let $a_1, \ldots, a_z$ ($z \geq 2$) be the lengths of the cyclic runs of ones between consecutive zeros. If all a_i were odd, then $n = z + \sum a_i \equiv 2z \equiv 0 \pmod 2$, which is false; so some a_i is even. The rotation $w = 0\,1^{a_i}\,0\cdots$ has prefix parities equal to those of E_0 up to position $a_i + 1$, where both are 0; at position $a_i + 2 \leq n$ the parity of w stays 0 while that of E_0 becomes 1. So $\mathrm{rk}(w) < \mathrm{rk}(E_0)$, and the first occurrence of X precedes that of $\langle E_0\rangle$. ∎

Lemma 3: $\langle A_k\rangle$ and $\langle E_k\rangle$ are adjacent in Q_n for $0 \leq k \leq K-1$.

Proof: Flipping position $n-3$ of E_0 gives A_0. For $k \geq 1$, the end before step k is (a rotation of) $0^{2k-1}10\,1^{c_k}01$; its rotation $01\,0^{2k-1}10\,1^{c_k}$ becomes A_k when its second bit is flipped. (For $k = 1$ this word is a rotation of E_1 as defined above.) ∎

Lemma 4: A_k is the first occurrence of its necklace.

Proof: The prefix parity $\pi(y)$ begins with exactly as many zeros as y. For $k \geq 1$, A_k begins with 0^{2k+1} and ends with 1, its cyclic runs of zeros have lengths $2k+1$ and 1, and $2k+1 \geq 3$; so A_k is the only rotation whose prefix parity begins with $2k+1$ zeros, and it has the smallest rank. For $k = 0$ write $A_0 = 01^a0111$ with $a = n-5 \geq 2$ even. Rotations beginning with 1 are larger, and the only other rotation beginning with 0 is $R = 01110\,1^a$. The prefix parities of A_0 and R agree in positions 1–3; if $a = 2$, position 4 gives 0 for A_0 and 1 for R, and if $a \geq 4$ they agree in position 4 and position 5 gives 0 for A_0 and 1 for R. ∎

Lemma 5: $g(\mathrm{rk}(A_k)+1) = E_{k+1}$, and E_{k+1} is the first occurrence of its necklace. Hence $\langle E_{k+1}\rangle$ immediately follows $\langle A_k\rangle$ in D_n.

Proof: The words $g(r)$ and $g(r+1)$ differ exactly in the position of the lowest set bit of $r+1$. The word A_k ends in an even block of ones preceded by a zero (for $k \geq 1$ the block 1^{c_k}, entered with parity 1; for $k = 0$ the block 111, entered with parity 0), so $\pi(A_k)$ ends in 01; thus $r = \mathrm{rk}(A_k) \equiv 1 \pmod 4$, the lowest set bit of $r+1$ is in position $n-1$, and flipping position $n-1$ of A_k gives E_{k+1}. For $k \geq 1$, E_{k+1} begins with 0^{2k+1}, ends with 1, and its other runs of zeros have length 1 or 2, so the argument of Lemma 4 applies. For $k = 0$, the rotations of $E_1 = 01^a0101$ beginning with 0 are E_1, $01010\,1^a$ and $010\,1^a01$, whose prefix parities begin 010, 011 and 011. Since E_{k+1} is the BRGC successor of A_k and is a first occurrence, no necklace has its first occurrence strictly between them. ∎

Lemma 6: Let j_k be the position of $\langle A_k\rangle$ in D_n. Starting from D_n and performing the Pósa rotations at $\langle A_0\rangle, \langle A_1\rangle, \ldots, \langle A_{K-1}\rangle$ in turn is legal, and after the rotation at $\langle A_k\rangle$ the end of the path is $\langle E_{k+1}\rangle$.

Proof: By Lemma 5, $\langle E_{k+1}\rangle$ is at position $j_k + 1$. For $k = 0$: comparing prefix parities, $\mathrm{rk}(A_0) < \mathrm{rk}(E_1) < \mathrm{rk}(E_0)$ (the words agree up to position $n-4$, and at position $n-3$ the parity is 0 for E_1 and 1 for E_0), so $j_0 + 1$ is strictly before the end, $\langle A_0\rangle$ is neither the end nor its predecessor, and by Lemma 3 the rotation is legal with new end $\langle E_1\rangle$. For $k \geq 1$: A_k begins with exactly $2k+1$ zeros and A_{k-1} with $0^{2k-1}1$, so $\mathrm{rk}(E_{k+1}) = \mathrm{rk}(A_k) + 1 \leq 2^{n-2k-1} < 2^{n-2k} \leq \mathrm{rk}(A_{k-1})$, and since D_n is ordered by rank, $j_k + 1 < j_{k-1}$. After the rotation at $\langle A_{k-1}\rangle$ the path is $P_0 \cdots P_{j_{k-1}}$ followed by reversed material, with end $\langle E_k\rangle$; both $\langle A_k\rangle$ and $\langle E_{k+1}\rangle$ lie in the untouched prefix, at consecutive positions, so the rotation at $\langle A_k\rangle$ (legal by Lemma 3) has new end $\langle E_{k+1}\rangle$. ∎

Proof of Theorem A: By Lemma 2 and the hypothesis, D_n is a Hamilton path of Q_n from $\langle 0^{n-1}1\rangle$ to $\langle E_0\rangle$. By Lemma 6 the K rotations produce a Hamilton path from $\langle 0^{n-1}1\rangle$ (the start never moves) to $\langle E_K\rangle = \langle 0^{n-4}1001\rangle$, which is adjacent to $\langle 0^{n-1}1\rangle$. This is a Hamilton cycle; apply Lemma 1. ∎

IV. Computations

a) Codes for $5 \leq n \leq 31$ by search: We found Hamilton cycles in Q_n for all primes $5 \leq n \leq 31$ by Pósa rotation and extension [10], with Warnsdorff extension and a closing look-ahead, started from the Degni–Drisko list or from a greedy path. The search took under 2 seconds for $n = 23$, 3.5 and 12 minutes for two codes at $n = 29$, and 17 minutes and 20.6 GB of memory for $n = 31$. Each code was expanded to its full period and checked by at least two independent programs among a C bitset checker, a NumPy checker that reads only the track and the head positions, and, for $n = 23$, 29 and 31, two further independently written checkers.

b) Codes from Theorem A: The construction of Theorem A is algorithmic: the first occurrences can be enumerated in rank order with $O(n)$ work each, and the cycle is a concatenation of $K+1$ segments of D_n, traversed forwards or backwards. Our implementation enumerates D_n in parallel, checks Conjecture 1 along the way, assembles the cycle, and lifts it to a flip sequence of length $m = (2^n-2)/n$ together with the shift s. For $7 \le n \le 29$ the resulting codes pass the C bitset checker, the NumPy checker and the flip-sequence verifier described next. For $n = 37$ the code has $m = 3{,}714{,}566{,}310$ and $s = 2$; it was produced in 15 minutes on four cores. A separately written verifier regenerated all $2^{37}-2$ words from the flip sequence, marking them in a 2^{37}-bit table, and confirmed that every nonconstant word occurs exactly once, that the n passes chain with shift 2, and that the sequence closes.

c) The Degni–Drisko property at $n = 41, 43, 47$: By Theorem A, a code for prime n follows from Conjecture 1 at n. We verified it by streaming the BRGC ranks with constant memory per thread. At $n = 41$, D_{41} has $53{,}634{,}713{,}550$ necklaces, every consecutive pair is adjacent, and the list runs from $\langle 0^{40}1\rangle$ to $\langle 01^{40}\rangle$ (77 minutes on four cores). A second program, written independently without access to the first (it computes the ranks of all rotations incrementally and scans only ranks below 2^{n-1}), confirmed the same count, the same first and last necklaces and no non-adjacent pair (36 minutes on four cores); it also confirmed $n = 37$. At $n = 43$ ($204{,}560{,}302{,}842$ necklaces) both programs found every consecutive pair adjacent (4.7 and 3.0 hours on four cores, respectively). At $n = 47$ ($2{,}994{,}414{,}645{,}858$ necklaces) both programs, with checkpointing, again found every consecutive pair adjacent and the expected first and last necklaces (64 and 55 hours of wall time, respectively, on four cores of a shared laptop).

n	period 2^n-2	method	verification
3	6	explicit	Lean (`decide`)
5–19	$\le 524{,}286$	search; Theorem A ($n \ge 7$)	word by word, 2 checkers
23	8,388,606	search ($s = 1$, $s = 11$)	word by word, ≥ 3 checkers
29	536,870,910	search ($s = 12$, $s = 16$)	word by word, ≥ 3 checkers
31	2,147,483,646	search ($s = 28$)	word by word, ≥ 3 checkers
37	137,438,953,470	Theorem A ($s = 2$)	word by word, 1 checker
41	2,199,023,255,550	Theorem A	Conjecture 1 at $n = 41$, 2 programs
43	8,796,093,022,206	Theorem A	Conjecture 1 at $n = 43$, 2 programs
47	140,737,488,355,326	Theorem A	Conjecture 1 at $n = 47$, 2 programs

TABLE I

OPTIMAL SINGLE-TRACK GRAY CODES. "WORD BY WORD" MEANS THAT A PROGRAM WRITTEN SEPARATELY FROM THE GENERATOR REGENERATED EVERY WORD OF THE PERIOD. FOR $n = 41, 43, 47$ THE CODES ARE THE EXPLICIT ONES OF THEOREM A; THEY WERE NOT EXPANDED.

V. FURTHER OBSERVATIONS AND OPEN PROBLEMS

a) Symmetric chains: Griggs, Killian and Savage [4] (see also [7]) gave, for prime n, a rotation-invariant symmetric chain decomposition of the Boolean lattice; it induces a decomposition of Q_n into paths. Gregor, Mička and Mütze [5] proved that the Greene–Kleitman decomposition of the cube extends to a Hamilton cycle and conjectured that every symmetric chain decomposition does.

Conjecture 2: For every prime n the Griggs–Killian–Savage chains (with 0^n and 1^n removed) extend to a Hamilton cycle of Q_n.

We found such cycles for every prime $7 \le n \le 23$ by a search on the chains as two-ended super-vertices. Every chain has an odd number of edges, so chain bottoms are joined only to bottoms and tops only to tops, apart from joins between chains of the middle two levels; these cycles are "cycle orderings" in the sense of [5].

b) Central levels: Merino, Mička and Mütze [8] showed that the middle two levels of the cube have a Hamilton cycle invariant under rotation, for every odd n; in particular, the middle two levels of the necklace graph are Hamiltonian, for prime and composite odd n.

Conjecture 3: For every odd $n = 2m+1 \ge 5$ and every $1 \le \ell \le m$, the subgraph of the necklace graph induced by the necklaces of weight in $[m+1-\ell, m+\ell]$ is Hamiltonian.

The case $\ell = 1$ is the theorem of [8], and the case $\ell = m$, n prime, is equivalent (by Lemma 1) to the existence of optimal single-track Gray codes. Our search found such Hamilton cycles, checked by the search program itself, for all odd $n \le 19$ and all ℓ. For $\ell = m$ and $n = 9$, 13, 15, 21, 25 (periodic necklaces included for composite n) the cycles also passed an independent checker.

c) Structure of the Degni–Drisko conjecture: Degni and Drisko prove that the strings skipped between consecutive Gray necklaces are covered by their "symmetric intervals", note that removing a single such interval preserves Hamming distance one, and identify several intervals as the difficulty [1, Section 8]. For prime $n \leq 13$ every gap between consecutive necklaces of D_n is a single symmetric interval. This fails at $n = 17$ and $n = 19$: in all 7 and 21 exceptional gaps, respectively, the gap consists of exactly three symmetric intervals, a central one flanked by two whose generators have equal length, between $n - 5$ and $n - 3$.

SUPPLEMENTARY MATERIAL

The supplementary material (also posted as ancillary files with the arXiv version) contains the Lean 4 project (`lean/`, with toolchain and Mathlib pins), the programs for the search, the construction of Theorem A and the verifiers, the tracks and head positions of the search codes for $5 \leq n \leq 23$ (the tracks for $n = 29$ and 31, 0.5 and 2.1 GB uncompressed, are available at https://github.com/dzoba/single-track-gray-codes), and the logs of the Degni–Drisko verifications at $n = 37$, 41, 43 and 47 and of the checks described in Section IV. The $n = 37$ flip sequence (3.7 GB, SHA-256 `0a0c91ea19527ee4812ff2b0d8627d3abe6e0d4fbaad611a48efbb13f8c5b811`) can be regenerated by the included construction program.

ACKNOWLEDGMENT

This work was carried out with AI systems under the author's direction. Claude (Anthropic) wrote and ran the searches and verifiers, found the closing pivots of Theorem A by computer search, wrote the proofs and the Lean formalization, and drafted the text. The word-by-word verifiers share no code with the generators, and the second Degni–Drisko checker was written by a separate AI instance from a specification only. Separate instances refereed the proof and the formalization, and Codex (OpenAI) wrote two further checkers and checked the codes for $n = 23$, 29 and 31. The author is responsible for the content.

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